

ELECTRONIC DEVICES AND CIRCUIT THEORY

TENTH EDITION

BOYLESTAD



PEARSON

Chapter 6: Field-Effect Transistors

FETs vs. BJTs

Similarities:

- Amplifiers
- Switching devices
- Impedance matching circuits

Differences:

- FETs are voltage controlled devices. BJTs are current controlled devices.
- FETs have a higher input impedance. BJTs have higher gains.
- FETs are less sensitive to temperature variations and are more easily integrated on ICs.
- FETs are generally more static sensitive than BJTs.

FET Types

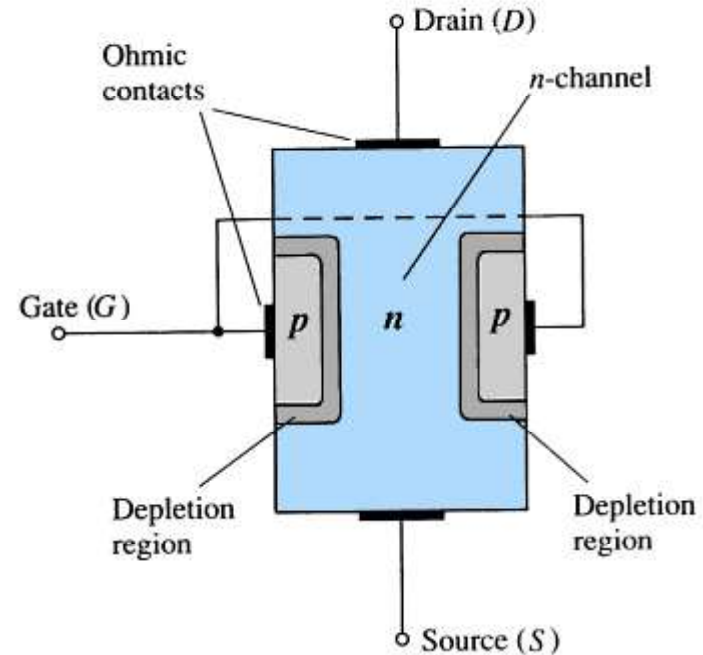
- **JFET:** Junction FET
- **MOSFET:** Metal–Oxide–Semiconductor FET
 - **D-MOSFET:** Depletion MOSFET
 - **E-MOSFET:** Enhancement MOSFET

JFET Construction

There are two types of JFETs

- ***n*-channel**
- ***p*-channel**

The ***n*-channel** is more widely used.



There are three terminals:

- **Drain (D)** and **Source (S)** are connected to the *n*-channel
- **Gate (G)** is connected to the *p*-type material

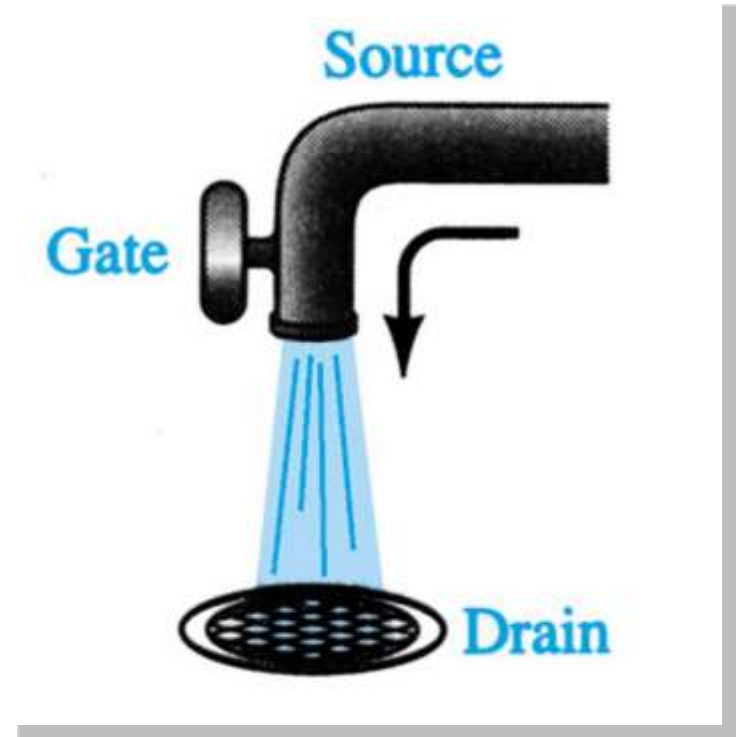
JFET Operation: The Basic Idea

JFET operation can be compared to a water spigot.

The source of water pressure is the accumulation of electrons at the negative pole of the drain-source voltage.

The drain of water is the electron deficiency (or holes) at the positive pole of the applied voltage.

The control of flow of water is the gate voltage that controls the width of the n-channel and, therefore, the flow of charges from source to drain.



JFET Operating Characteristics

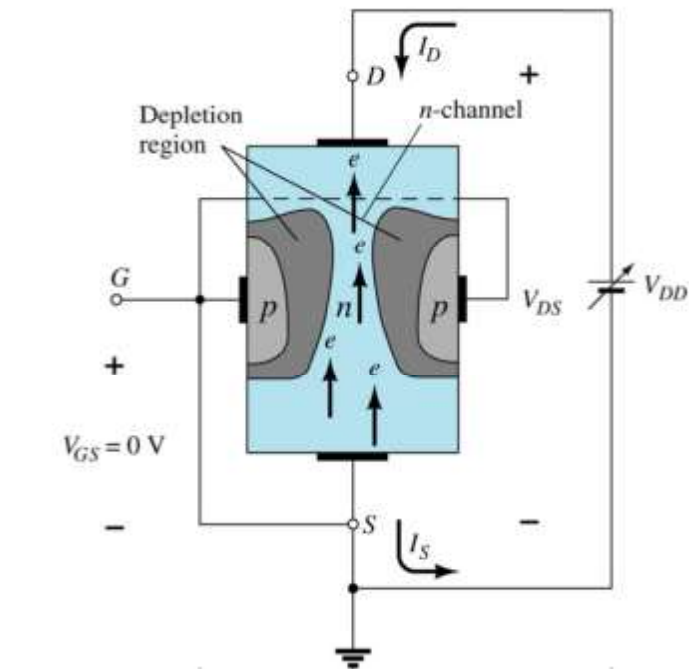
There are three basic operating conditions for a JFET:

- **$V_{GS} = 0$, V_{DS} increasing to some positive value**
- **$V_{GS} < 0$, V_{DS} at some positive value**
- **Voltage-controlled resistor**

JFET Operating Characteristics: $V_{GS} = 0 \text{ V}$

Three things happen when $V_{GS} = 0$ and V_{DS} is increased from 0 to a more positive voltage

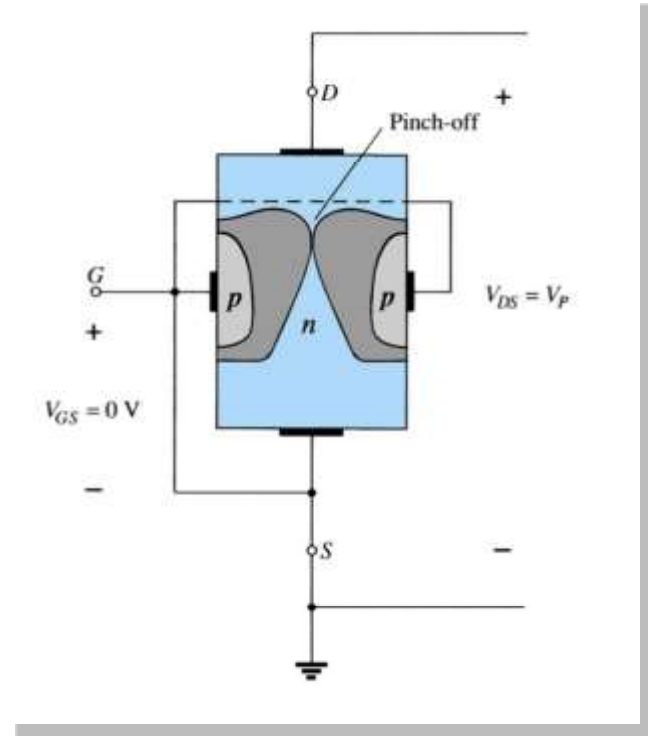
- The depletion region between p-gate and n-channel increases as electrons from n-channel combine with holes from p-gate.
- Increasing the depletion region, decreases the size of the n-channel which increases the resistance of the n-channel.
- Even though the n-channel resistance is increasing, the current (I_D) from source to drain through the n-channel is increasing. This is because V_{DS} is increasing.



JFET Operating Characteristics: Pinch Off

If $V_{GS} = 0$ and V_{DS} is further increased to a more positive voltage, then the depletion zone gets so large that it **pinches off** the n-channel.

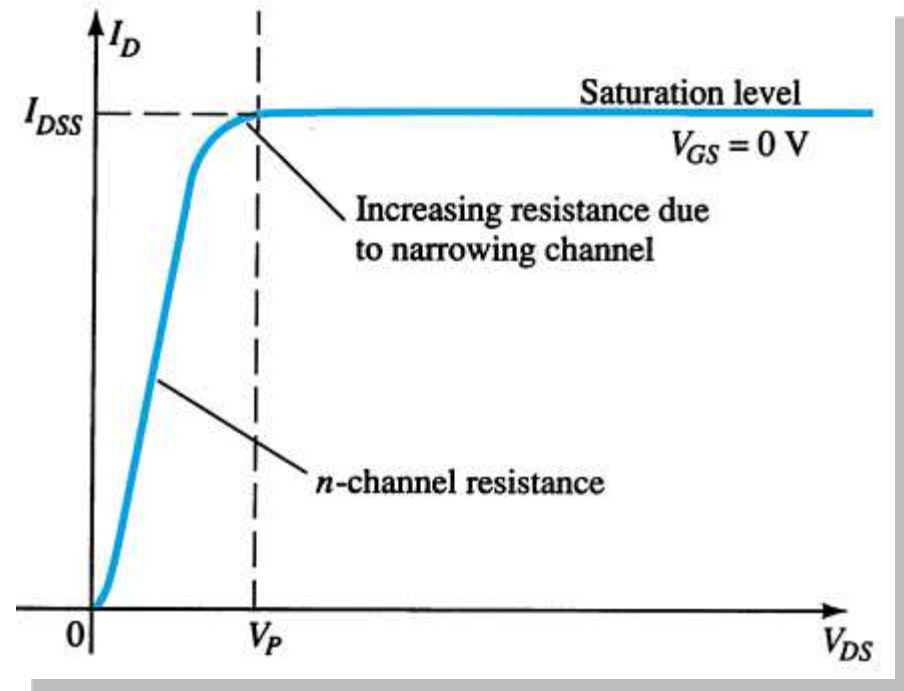
This suggests that the current in the n-channel (I_D) would drop to 0A, but it does just the opposite—as V_{DS} increases, so does I_D .



JFET Operating Characteristics: Saturation

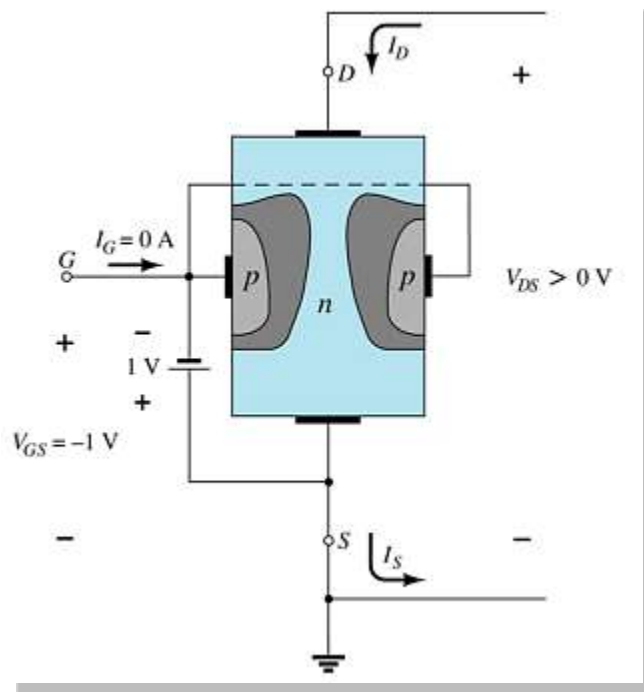
At the pinch-off point:

- Any further increase in V_{GS} does not produce any increase in I_D . V_{GS} at pinch-off is denoted as V_p .
- I_D is at saturation or maximum. It is referred to as I_{DSS} .
- The ohmic value of the channel is maximum.



JFET Operating Characteristics

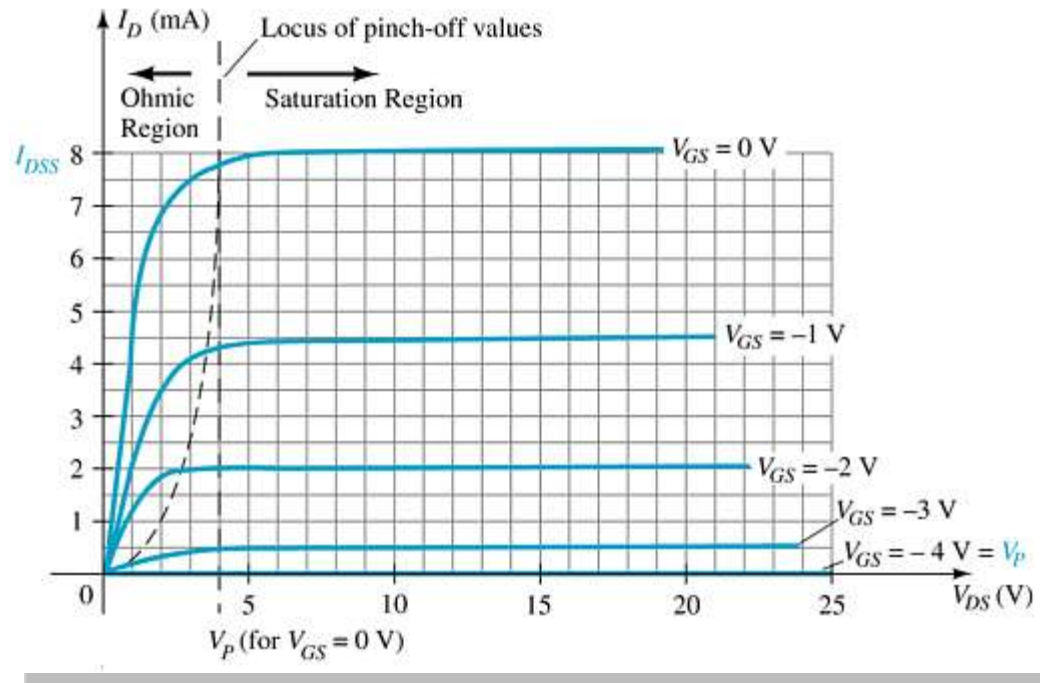
As V_{GS} becomes more negative, the depletion region increases.



JFET Operating Characteristics

As V_{GS} becomes more negative:

- The JFET experiences pinch-off at a lower voltage (V_P).
- I_D decreases ($I_D < I_{DSS}$) even though V_{DS} is increased.
- Eventually I_D reaches 0 A. V_{GS} at this point is called V_p or $V_{GS(off)}$.



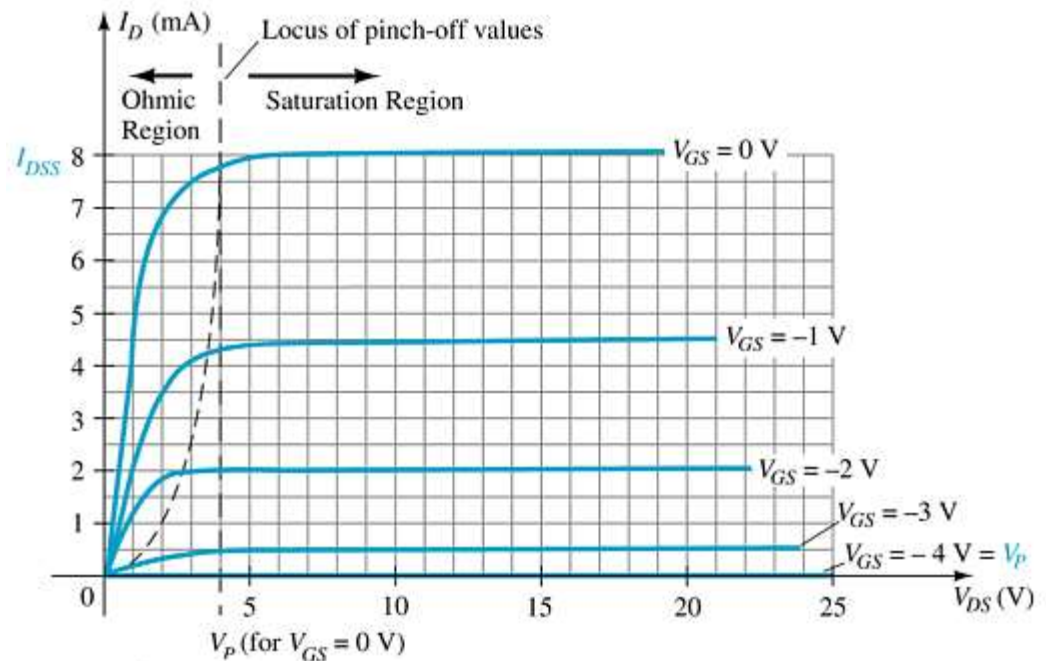
Also note that at high levels of V_{DS} the JFET reaches a breakdown situation. I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

JFET Operating Characteristics: Voltage-Controlled Resistor

The region to the left of the pinch-off point is called the **ohmic region**.

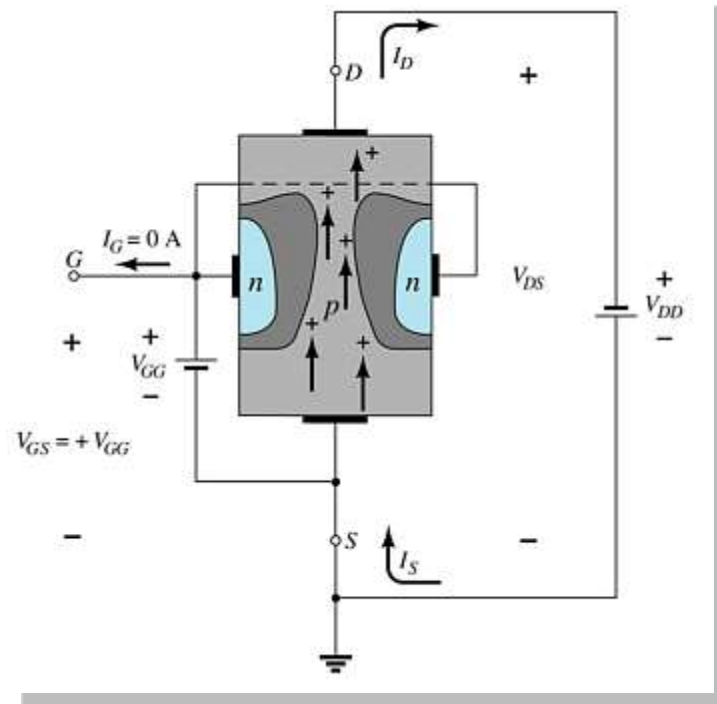
The JFET can be used as a variable resistor, where V_{GS} controls the drain-source resistance (r_d). As V_{GS} becomes more negative, the resistance (r_d) increases.

$$r_d = \frac{r_o}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$



p-Channel JFETs

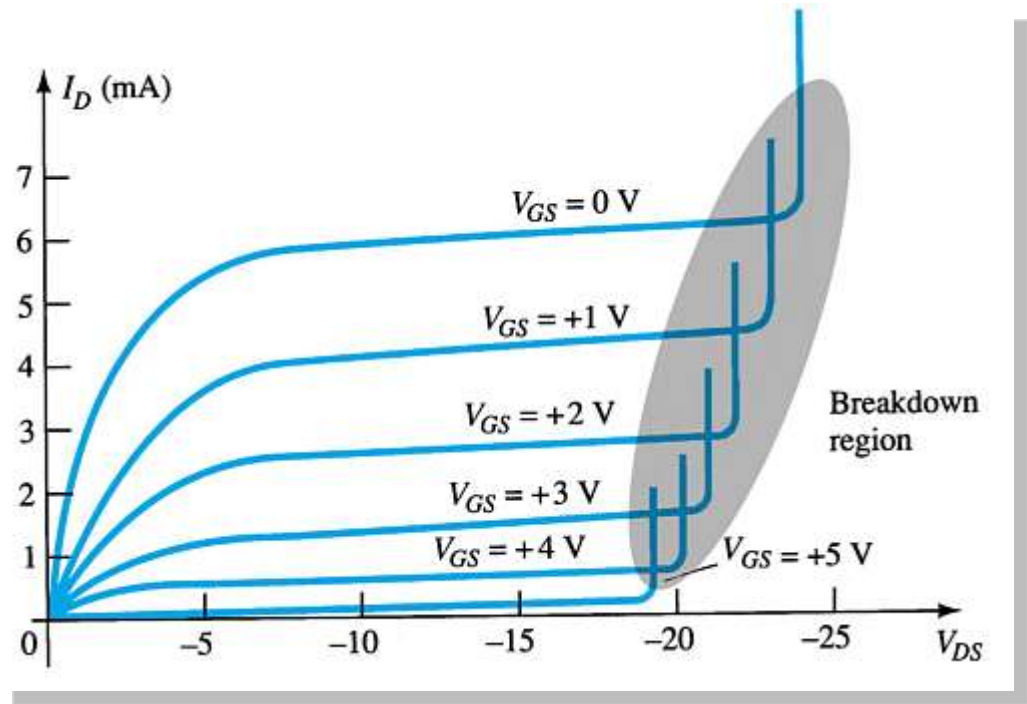
The *p*-channel JFET behaves the same as the *n*-channel JFET, except the voltage polarities and current directions are reversed.



p-Channel JFET Characteristics

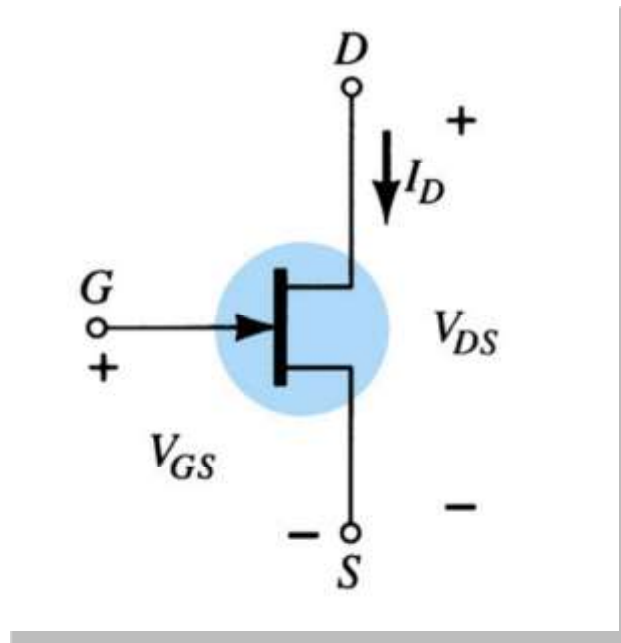
As V_{GS} increases more positively

- The depletion zone increases
- I_D decreases ($I_D < I_{DSS}$)
- Eventually $I_D = 0$ A



Also note that at high levels of V_{DS} the JFET reaches a breakdown situation: I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

N-Channel JFET Symbol



JFET Transfer Characteristics

The transfer characteristic of input-to-output is not as straightforward in a JFET as it is in a BJT.

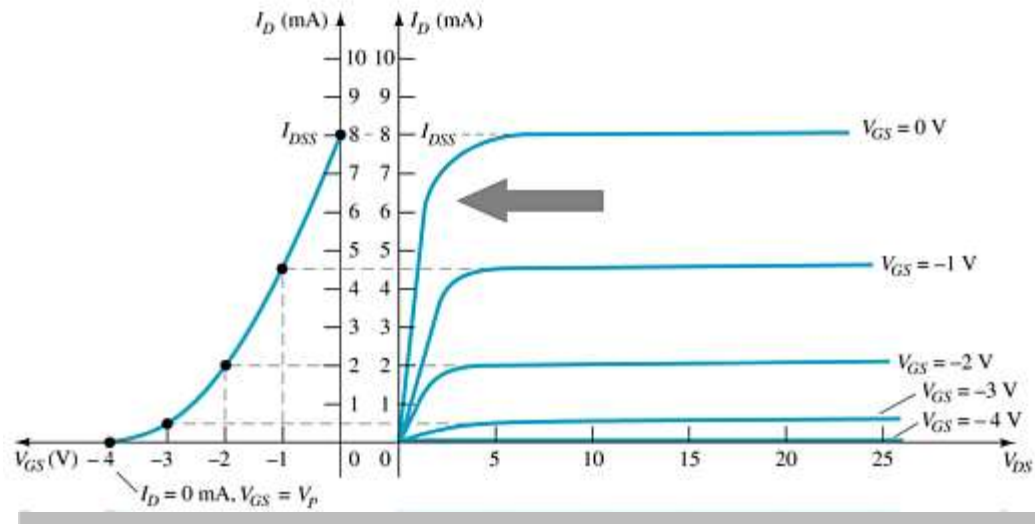
In a BJT, β indicates the relationship between I_B (input) and I_C (output).

In a JFET, the relationship of V_{GS} (input) and I_D (output) is a little more complicated:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

JFET Transfer Curve

This graph shows the value of I_D for a given value of V_{GS} .



Plotting the JFET Transfer Curve

Using I_{DSS} and V_p ($V_{GS(off)}$) values found in a specification sheet, the transfer curve can be plotted according to these three steps:

Step 1

Solving for $V_{GS} = 0V$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$
$$I_D = I_{DSS}$$

Step 2

Solving for $V_{GS} = V_p$ ($V_{GS(off)}$) $I_D = 0A$

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Step 3

Solving for $V_{GS} = 0V$ to V_p $I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$

MOSFETs

MOSFETs have characteristics similar to JFETs and additional characteristics that make them very useful.

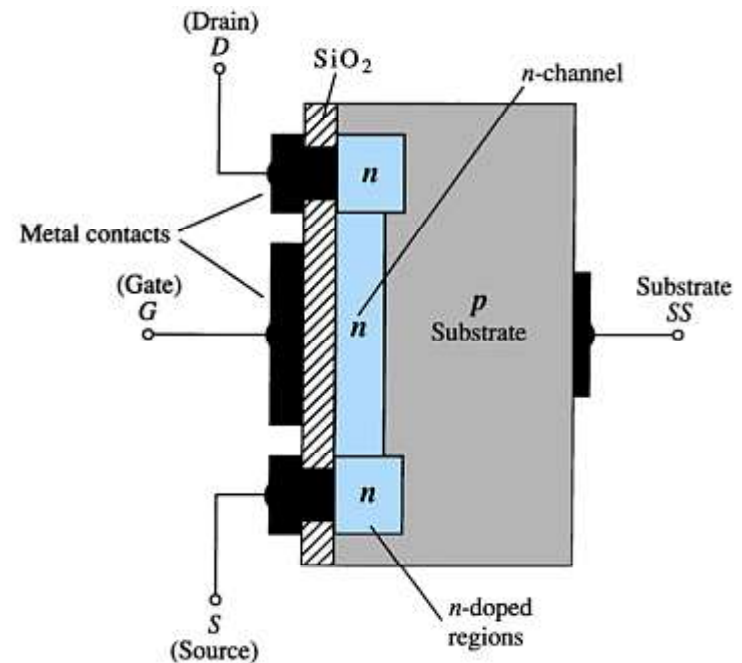
There are two types of MOSFETs:

- **Depletion-Type**
- **Enhancement-Type**

Depletion-Type MOSFET Construction

The **Drain** (D) and **Source** (S) connect to the n -doped regions. These n -doped regions are connected via an n -channel. This n -channel is connected to the **Gate** (G) via a thin insulating layer of SiO_2 .

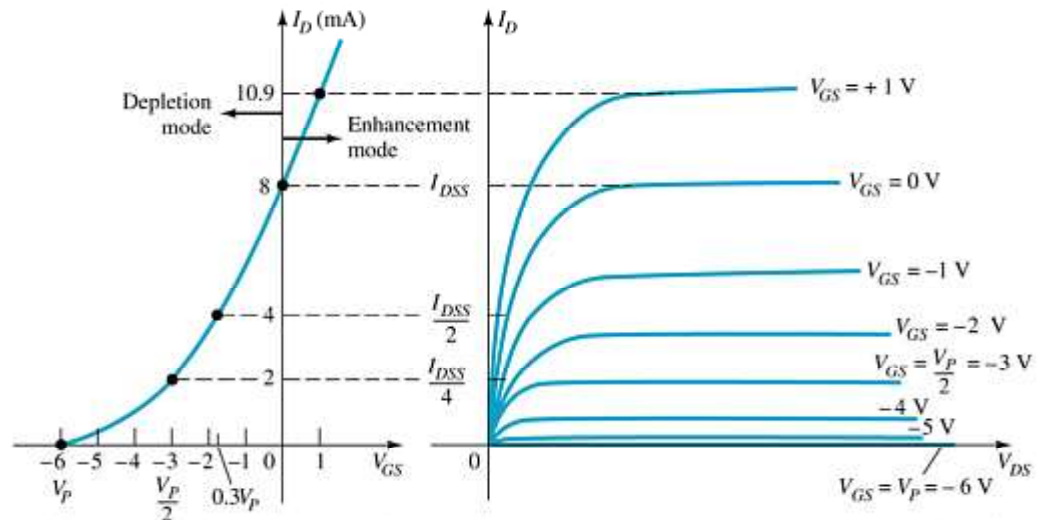
The n -doped material lies on a p -doped substrate that may have an additional terminal connection called **Substrate** (SS).



Basic MOSFET Operation

A depletion-type MOSFET can operate in two modes:

- Depletion mode
- Enhancement mode



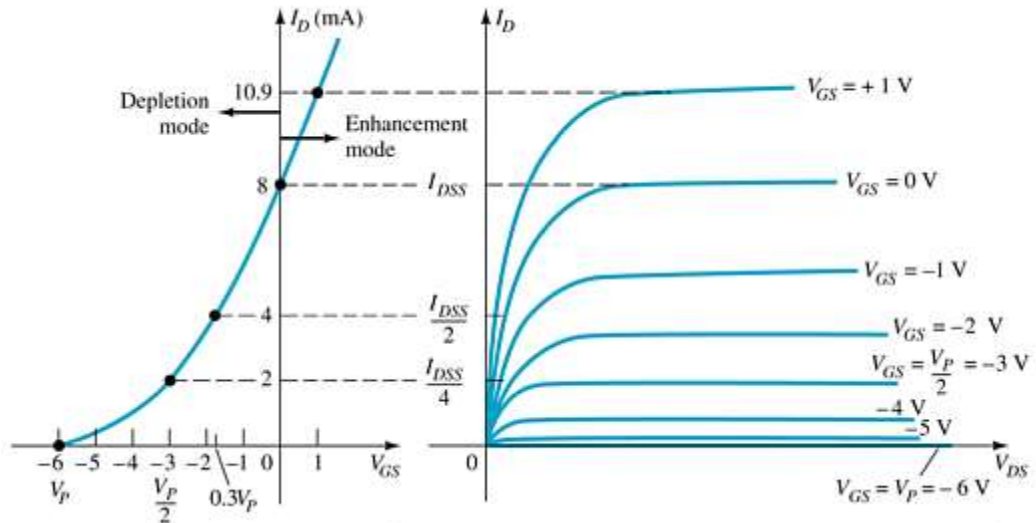
D-Type MOSFET in Depletion Mode

Depletion Mode

The characteristics are similar to a JFET.

- When $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS}$
- When $V_{GS} < 0 \text{ V}$, $I_D < I_{DSS}$
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

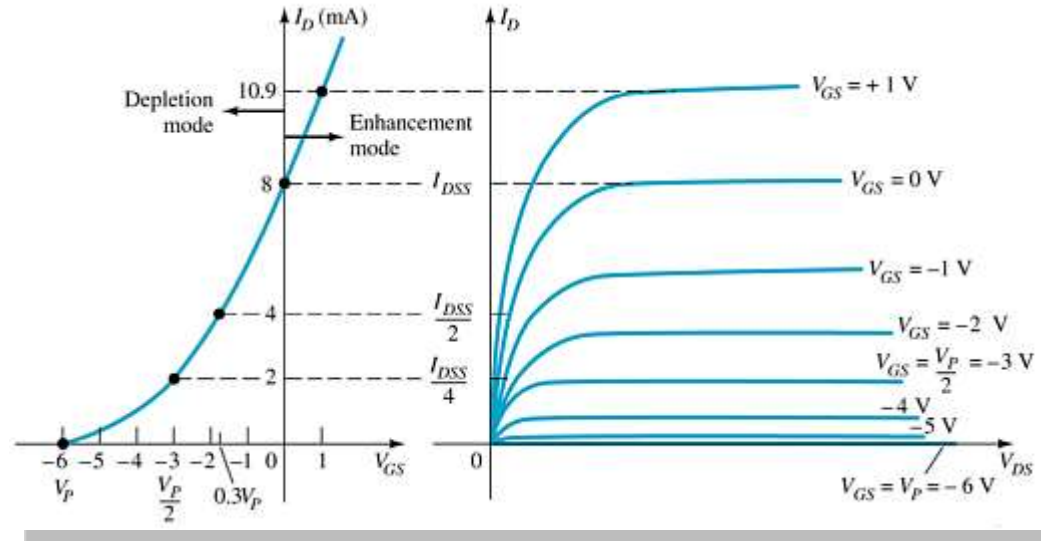


D-Type MOSFET in Enhancement Mode

Enhancement Mode

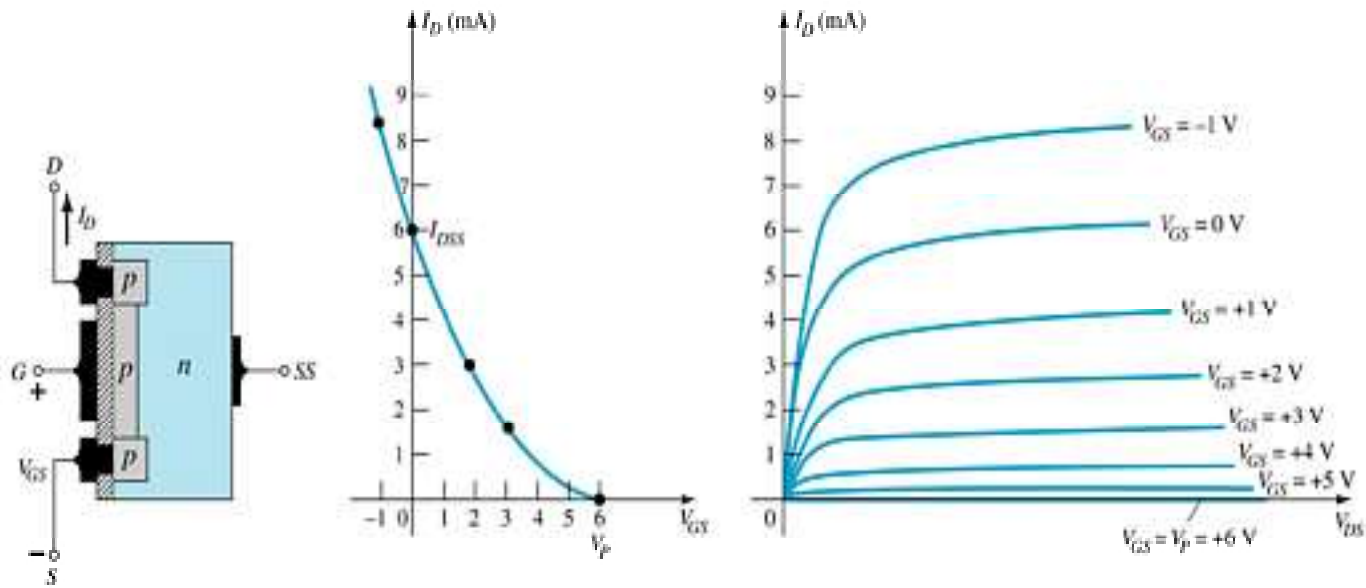
- $V_{GS} > 0 \text{ V}$
- I_D increases above I_{DSS}
- The formula used to plot the transfer curve still applies:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

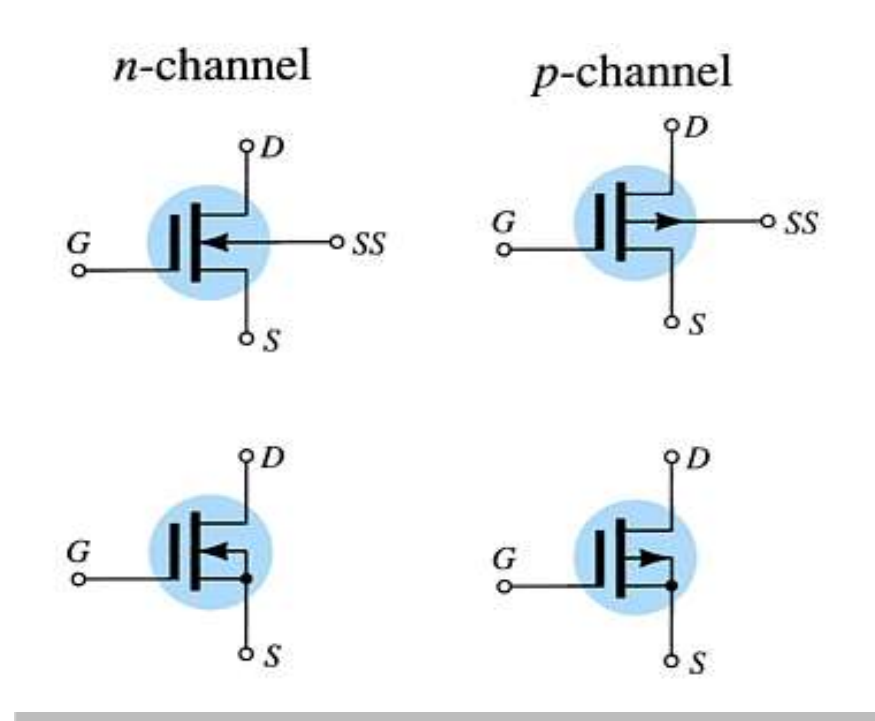


Note that V_{GS} is now a positive polarity

p-Channel D-Type MOSFET

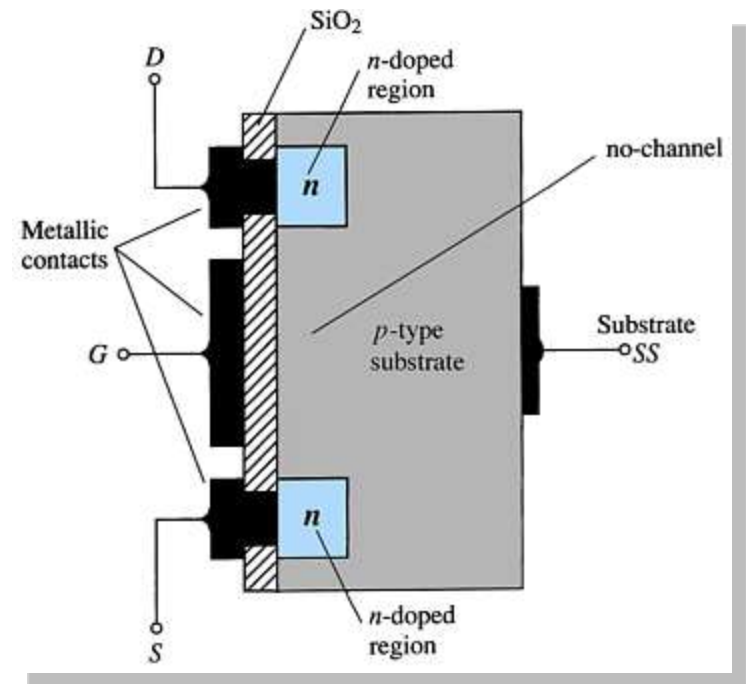


D-Type MOSFET Symbols



E-Type MOSFET Construction

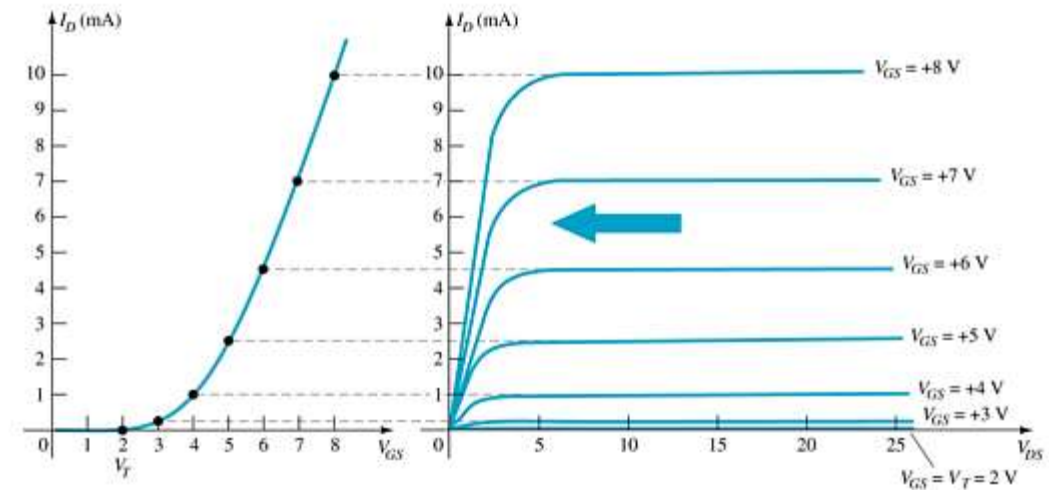
- The **Drain** (D) and **Source** (S) connect to the n -doped regions. These n -doped regions are connected via an n -channel
- The **Gate** (G) connects to the p -doped substrate via a thin insulating layer of SiO_2
- There is no channel
- The n -doped material lies on a p -doped substrate that may have an additional terminal connection called the **Substrate** (SS)



Basic Operation of the E-Type MOSFET

The enhancement-type MOSFET operates only in the enhancement mode.

- V_{GS} is always positive
- As V_{GS} increases, I_D increases
- As V_{GS} is kept constant and V_{DS} is increased, then I_D saturates (I_{DSS}) and the saturation level, V_{DSsat} is reached



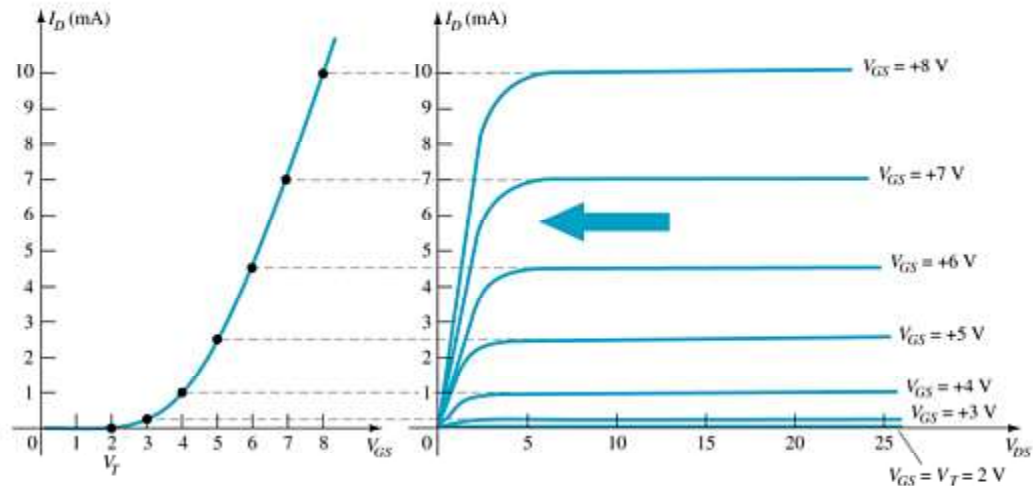
E-Type MOSFET Transfer Curve

To determine I_D given V_{GS} :

$$I_D = k(V_{GS} - V_T)^2$$

Where:

V_T = threshold voltage
or voltage at which the
MOSFET turns on



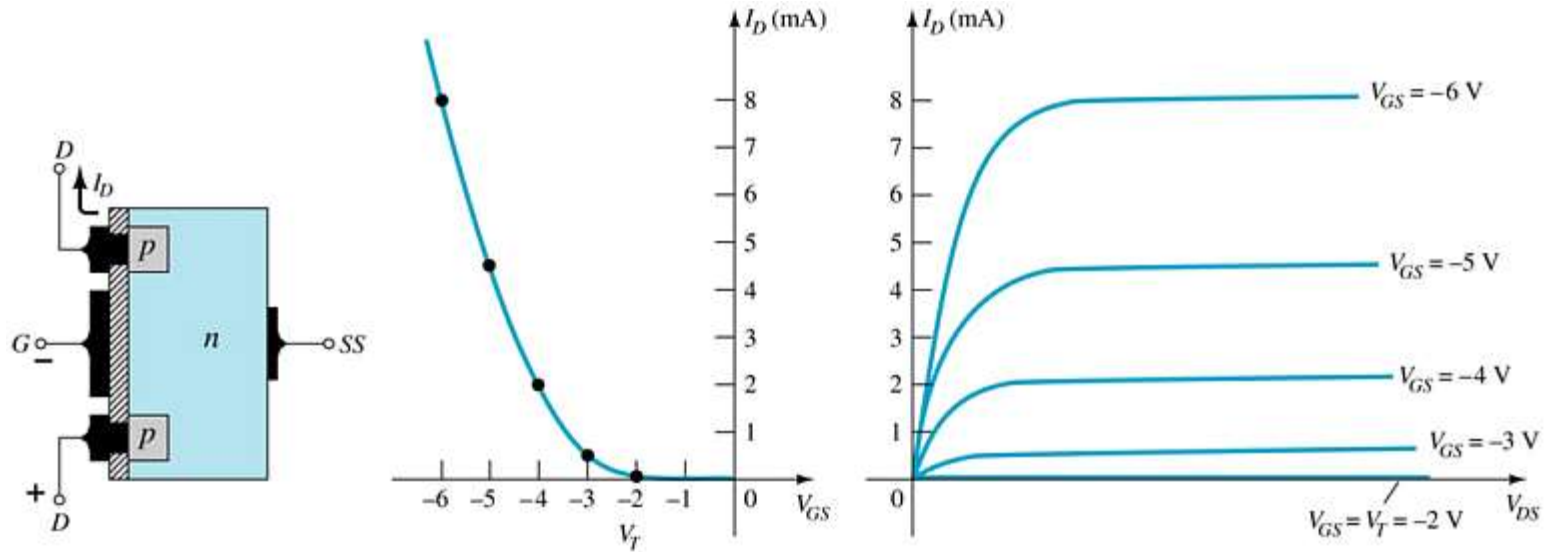
k , a constant, can be determined by using
values at a specific point and the formula:

$$k = \frac{I_{D(ON)}}{(V_{GS(ON)} - V_T)^2}$$

V_{DSsat} can be calculated by:

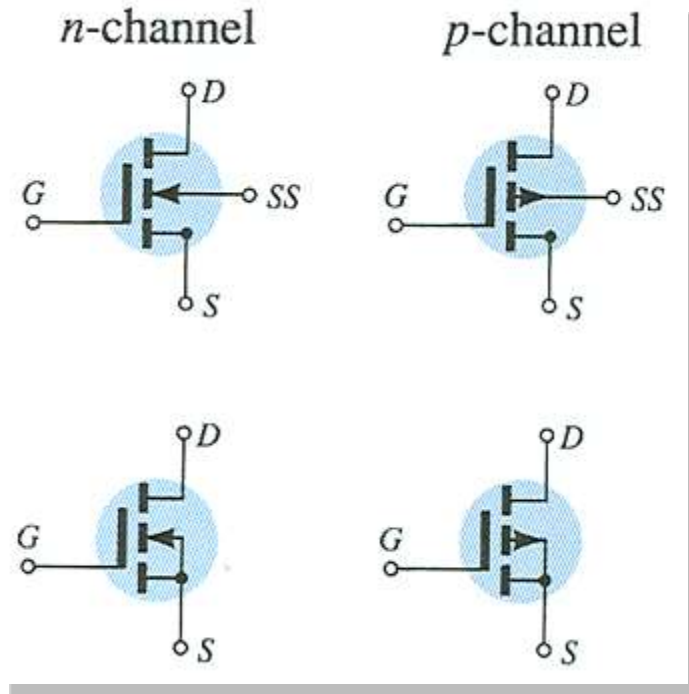
$$V_{DSat} = V_{GS} - V_T$$

p-Channel E-Type MOSFETs



The *p*-channel enhancement-type MOSFET is similar to the *n*-channel, except that the voltage polarities and current directions are reversed.

MOSFET Symbols



Summary Table

